

Isolation systems

Modern DevOps for Systems Biologists

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It works on my machine

Which is the problem

- ▶ TODO: the three things “my machine” is hiding
- ▶ TODO: why this bites hardest between teams, and across time

Environments

One language at a time

TODO: what an environment is in each of Python, Conda, Julia, R and Matlab, and what it is not.

```
python -m venv .venv && source .venv/bin/activate  
pip install -r requirements.txt
```

Lockfiles

TODO: the difference between what you asked for and what you got.

Dependency and security

Trusting other people's code

- ▶ TODO: the supply chain, stated without alarm
- ▶ TODO: what a version pin is actually protecting

Containers

Isolation one level down

TODO: what a container adds over an environment, and what it costs.

```
docker build -t analysis .  
docker run --rm -v "$PWD/data:/data" analysis
```

Documentation

The four kinds

TODO: tutorial, how-to, reference, explanation — and which one a `Quickstart.md` is.

Next

Activity 4

- ▶ TODO: what they will containerise
- ▶ TODO: what counts as done